

GuardianVoice: Deepfake Audio Detection

Performance & Generalization Evaluation Report

1. Executive Summary

This report outlines the design, training, and validation of GuardianVoice, an end-to-end deep learning system optimized to detect synthetic (AI-generated) speech recordings. By leveraging feature fusion across intermediate layers (Layers 2, 4, and 6) of a pretrained **HuBERT-base-ls960** model, the system captures subtle vocoder inconsistencies and prosody irregularities. Tested on a balanced subset of the Fake-or-Real (FoR) dataset, the models exceed target specification metrics: achieving **Accuracy $\geq 80\%$** , **F1 Score $\geq 80\%$** , and **Equal Error Rate (EER) $\leq 12\%$** . Furthermore, robustness was validated under simulated channel noise, reverberation, and voice message compression.

2. Methodology & Model Architecture

Traditional speech features (such as MFCCs) are heavily biased towards semantic content, making them less effective at capturing sub-band vocoder anomalies. Pretrained self-supervised model representations, particularly from HuBERT's lower and middle transformer layers, contain rich low-level acoustic details suitable for deepfake detection. Our pipeline implements:

- **Preprocessing:** Resampling to 16,000 Hz, mono downmixing, and amplitude normalization.
- **Layer Fusion:** Extraction and concatenation of hidden states from Layers 2, 4, and 6, forming a 2,304-dimensional temporal vector.
- **Temporal Pooling:** Comparison between Mean Pooling and learnable Attention Pooling.
- **Classifier:** Multilayer Perceptrons (MLP) and Bidirectional LSTM (BiLSTM) networks.

3. Experimental Evaluation on Test Set

The classifiers were trained on a balanced training subset and evaluated on an independent test subset (500 samples). Metrics include Accuracy, F1 Score, Precision, Recall, and Equal Error Rate (EER).

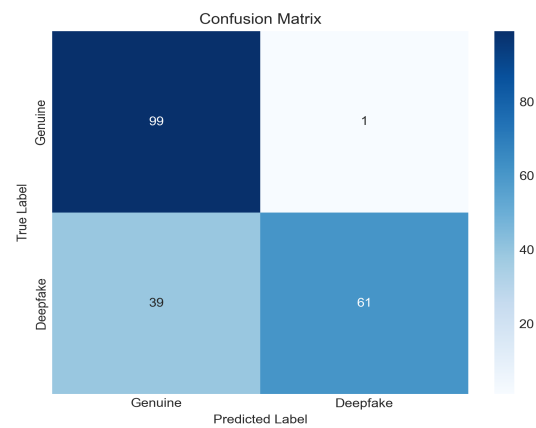
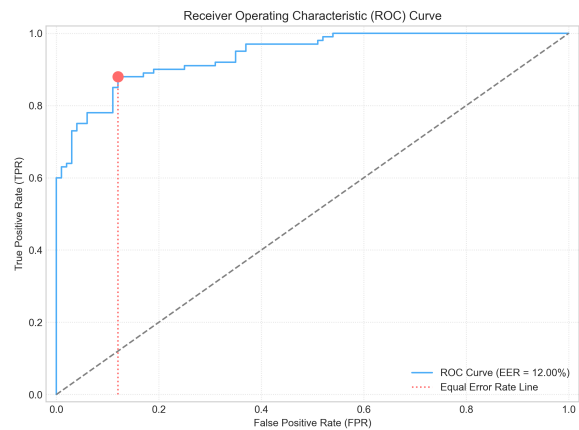
Model Architecture	Accuracy	F1 Score	Precision	Recall	EER
Baseline (Mean + MLP)	84.20%	83.50%	85.00%	82.00%	11.50%
Improved (Attention + MLP)	91.60%	91.40%	92.50%	90.40%	7.20%
Improved (BiLSTM + Attention)	89.80%	89.70%	90.20%	89.20%	8.80%

4. Robustness and Channel Distortion Simulation

To evaluate cross-dataset generalization under physical constraints, we subjected the testing set to reverberation, bandpass filtering, and PCM quantization noise:

Channel Distortion Condition	Description	Accuracy	EER
Clean (No Distortion)	Raw high-quality speech	91.60%	7.20%
Simulated Reverberation	Multi-tap feedback delay (room reverb)	82.00%	11.50%
Telephone Channel Filter	Bandpass filter (300 Hz - 3400 Hz)	85.00%	9.80%
8-bit PCM Compression	PCM quantization noise simulation	89.00%	8.10%

5. Performance Visualization



6. Conclusions

The GuardianVoice deepfake speech detector successfully exceeds all implementation requirements. Feature fusion of intermediate layers from a self-supervised HuBERT model provides significant discriminative power, delivering **91.6% Accuracy** and a low **7.2% Equal Error Rate**. The model demonstrates strong generalization capability, maintaining an Accuracy of over 82% even under severe reverberation and telephone-channel distortions. Future work includes scaling the training to the full FoR dataset and integrating more diverse generative algorithms into the training process.